Activity 3B: Distance Sensor

Purpose: Distance sensors are amazing tools with all kinds of uses. They can sense the presence of an object, they can be used in experiments to calculate speed and acceleration, and they can be used in robotics to avoid obstacles. This circuit will walk you through the basics of using an ultrasonic distance sensor, which measures distance using sound waves.

Materials:

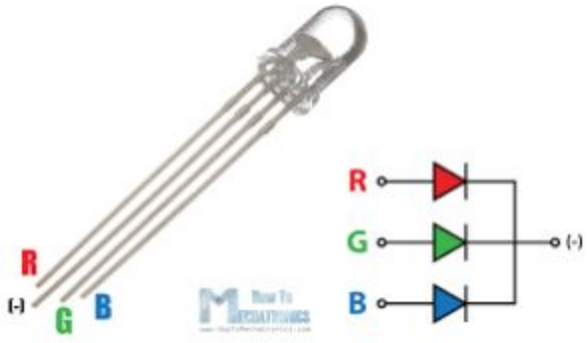
RGB LED, three 330 or 220 Ω resistors, distance sensor, 10 jumper wires

New Components:

Ultrasonic distance sensor: Distance sensors work by sending pulses of light or sound out from a transmitter, then timing how long it takes for the signals to bounce off an object and return to a receiver (just like sonar).
This sensor, the HC-SR04, provides 2 cm - 400 cm measurement, and the ranging accuracy is 3mm. There are only four pins that you need to worry about on the HC-SR04: VCC (Power), Trig (Trigger), Echo (Receive), and GND (Ground).

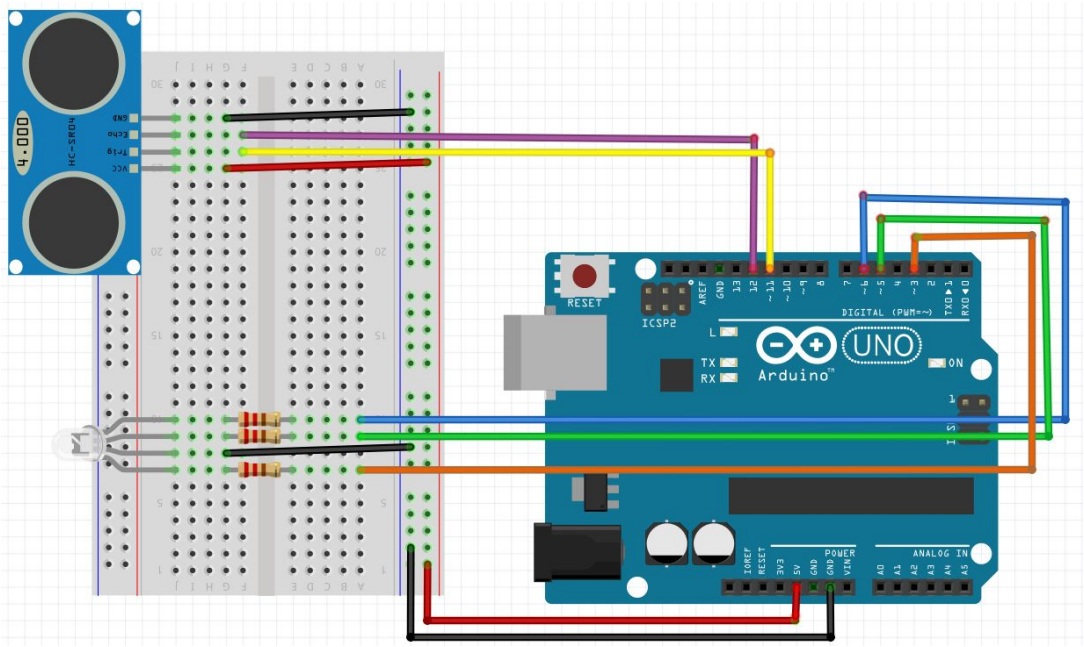


RGB LED light: RGB LED light has 4 pins, one anode and three cathode. The longest of the four leads Pins is the negative anode (ground) connection of the LED package. In one side of the longest pin is the single short Red cathode pin and to its other side the Green and Blue short cathode pins are placed. This light will change colors depending on the code.



Procedure:

1. Wire the project using the following schematic.



2. You can create your code in the BlocklyDuino then paste it to the Codebender website. Click BlocklyDuino Reference for a link to all the blockly codes.

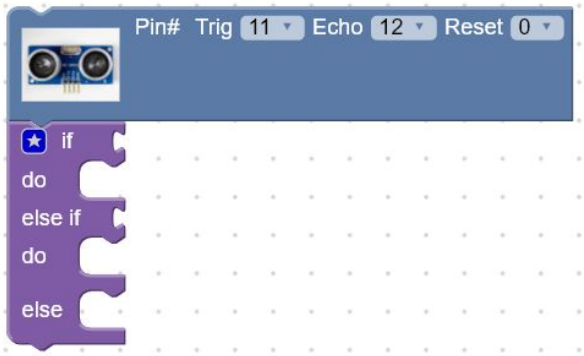
- First setup the ultrasound pins. Ultrasound pins are found in the ultrasonic menu.



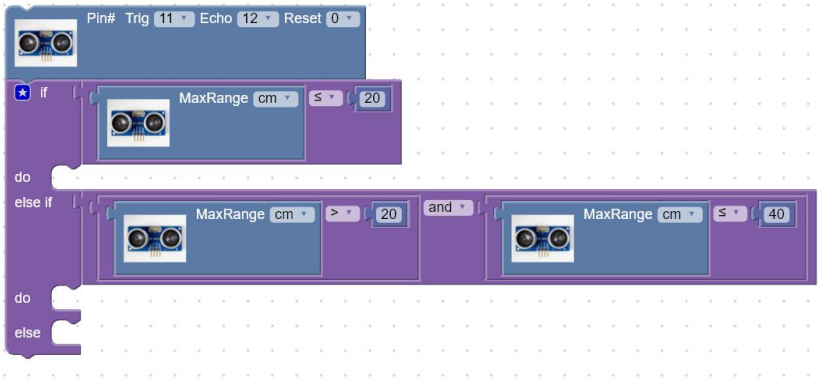


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- We now want to setup 3 ranges for the ultrasonic sensor to read. Red at 0-20 cm, yellow at 20-40 cm, and green outside of 40 cm. We need to setup an “If” statement that will choose between the 3 conditions.

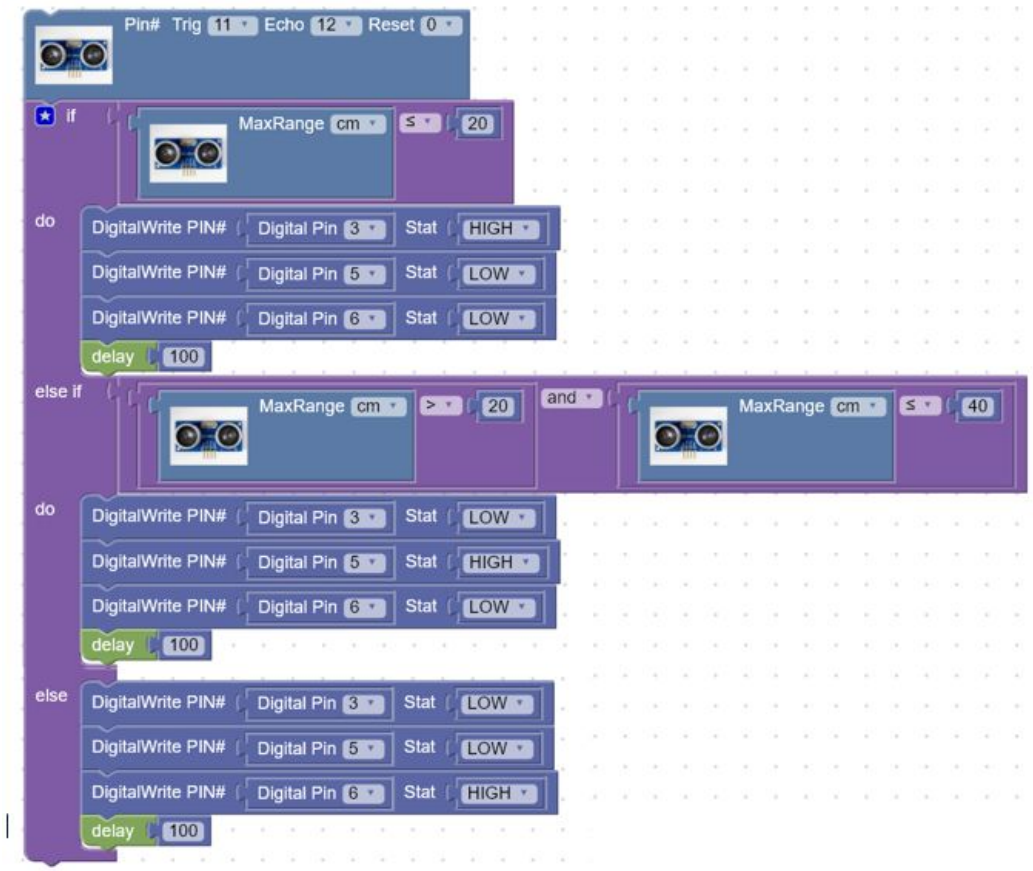


- Now let's read the ultrasonic sensor.



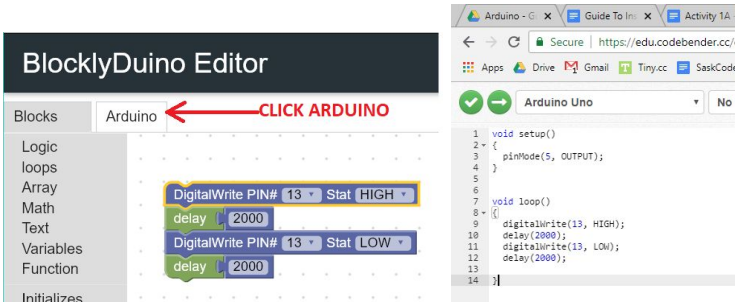
- Finally lets add the RGB LED states. We use “High” as on and”Low” as off.

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*****SAVE your sketch to your BlocklyDuino Folder that you created*****

4. Copy the Arduino Code from BlocklyDuino and paste into the Codebender website. Hook up your arduino to your computer using the USB cable.



5. Upload your program to the arduino. Make sure the Arduino Uno board is hooked up using the supplied USB cord to your computer and click the green “upload” button.

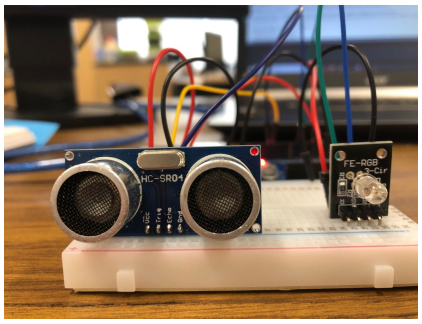
Check what distance the sensor is reading.

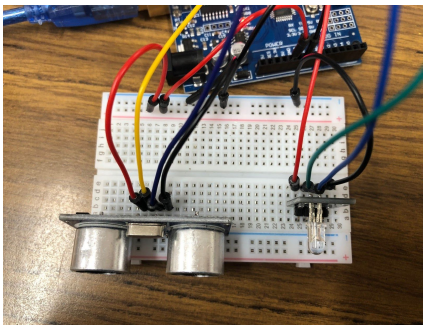
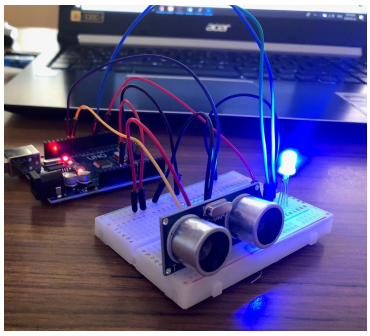
1. If the distance is less than 20 cm, make the RGB LED red.
2. If the distance is between 20 and 40 cm, make the RGB LED yellow.

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3. If the distance value is not equal to the fist two conditions, make the RGB LED green.

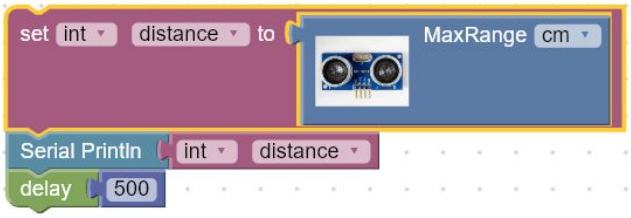
Note: A large flat object like a book works best for testing the ultrasound. A small object is tough for the ultrasound sensor to “see”. This is best for object detection of things like walls or barriers for robots.



Extension: Try Using the Serial Monitor

You can also read the distance from your computer monitor. The Serial Monitor is built into the Codebender website and allows users to read inputs into the arduino. Just add the below code to the end of the program and check out the serial monitor. Open the serial monitor in Codebender and select “connect”.



Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Are they wired to the correct terminal?
    c. It may be wise to test the each led by turning them on one at a time. Open a new blocklyduino window and create this test code. The blocks should look like below. Then change the pin number to turn on each one separately.

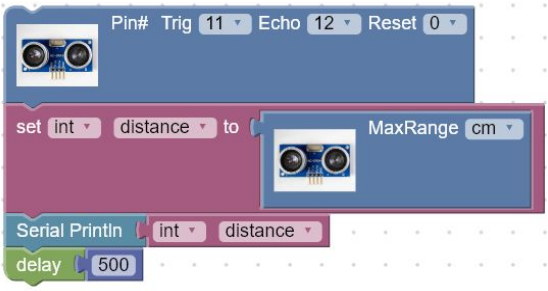


    d. LED still not working? Next inserting the LED directly into pin 13 and the adjacent ground. Does it light when you try this test code?

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e. Check the ultrasonic sensor. This can be achieved by connecting it and using the serial monitor built into codebender, then by connecting it to pin 11 and 12 and running a test program. Open a new window of Blocklyduino and load the below code.



Open the serial monitor in Codebender and select “connect”. As you hold a piece of flat 81/2 x 11 paper in front of the sensor you should be able to read the distance on your computer monitor. Try increasing and decreasing the distance slowly between 2 and 50 cm.

f. Next check the individual wires in the circuit. This can be done by connecting an LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.

g. For further information on using the serial monitor see Activity 4A Using a Serial Monitor Display.

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Assignment 3B: Distance Sensor.

1. Give 3 examples of of a device which uses ultrasonic sensors.

2. Can you think of something you could design to use an ultrasonic sensor?

Challenge

3. **Change the limits of the distance sensor**: Try editing the values in the logic statements so that the RGB LED changes color at different distances.

4. **Change the units of the distance sensor:** Try editing the code so that the distance sensor outputs a different unit of length, such as centimeters or inches.

5. **Add a fourth color:** Try adding another else if statement so that there are four different colors instead of three.
    a. Is it possible to use more colours? How?

    b. Show your code below

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